

Task Management REST API – Documentation

Project Name: Sentra World – Task Management

Base URL: http://localhost:8080

Authentication: API Key (Header-based)

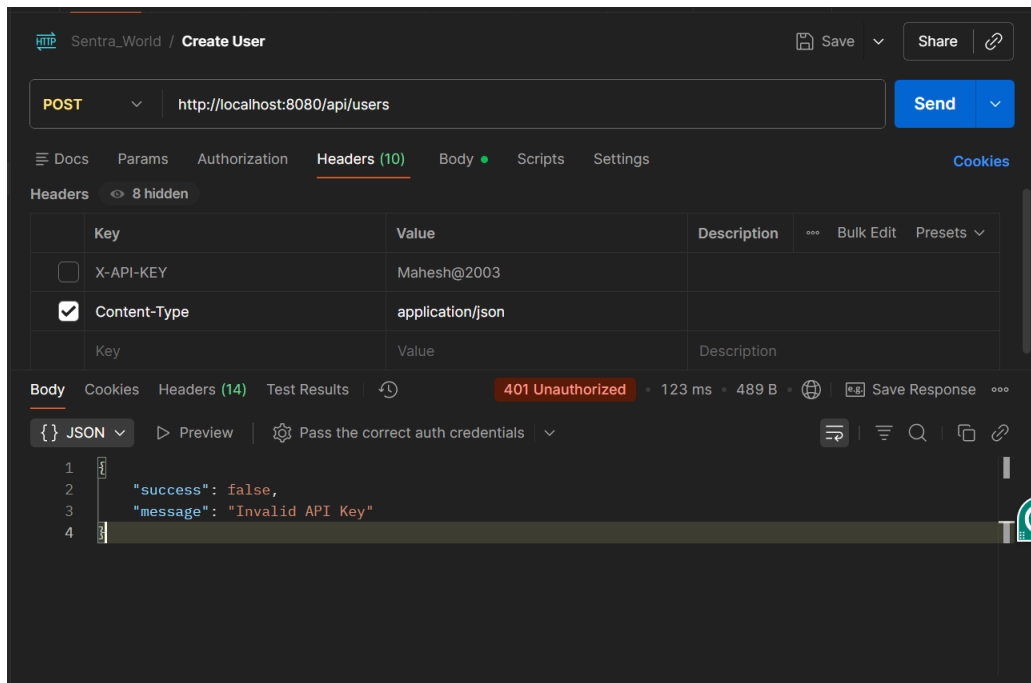
Header Name : X-API-KEY

Value : Mahesh@2003

If the API key is missing or invalid, the server returns:

```
{
  "success": false,
  "message": "Invalid API Key"
}
```

Screenshot:



User APIs

1. Create User

Endpoint:

POST /api/users

Request Body:

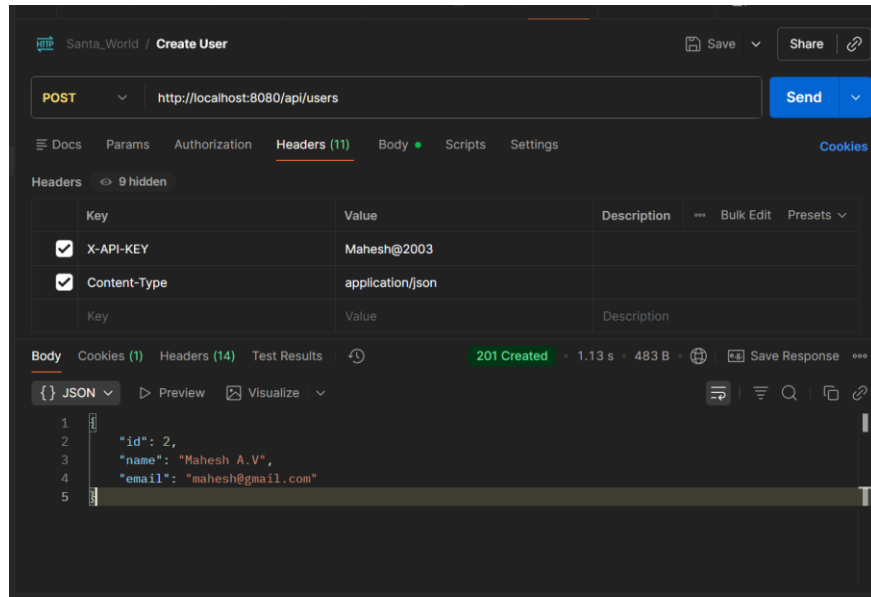
```
{
  "name": "Mahesh A.V",
  "email": "mahesh@gmail.com"
}
```

Success Response (201):

```
{
```

```
"id": 1,
"name": "Mahesh A.V",
"email": "mahesh@gmail.com"
}
```

Screenshot:



2. List Users (pagination)

Endpoint:

GET /api/users

Response:

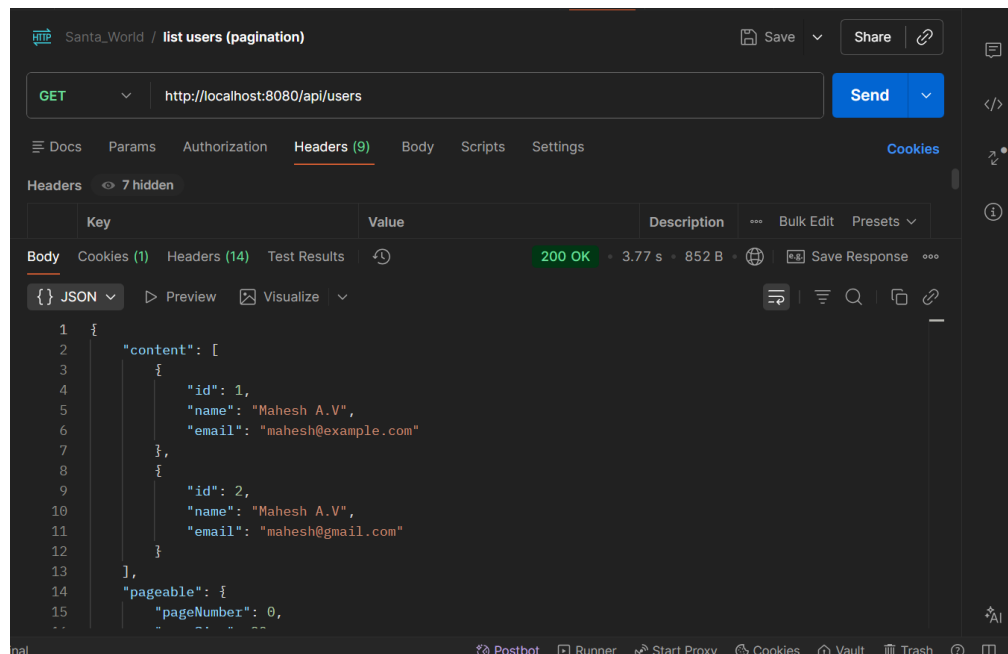
```
{
  "content": [
    {
      "id": 1,
      "name": "Mahesh A.V",
      "email": "mahesh@example.com"
    },
    {
      "id": 2,
      "name": "Mahesh A.V",
      "email": "mahesh@gmail.com"
    }
  ],
  "pageable": {
    "pageNumber": 0,
    "pageSize": 20,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "offset": 0,
    "paged": true,
  }
}
```

```

    "unpaged": false
  },
  "totalPages": 1,
  "totalElements": 2,
  "last": true,
  "size": 20,
  "number": 0,
  "sort": {
    "empty": true,
    "sorted": false,
    "unsorted": true
  },
  "first": true,
  "numberOfElements": 2,
  "empty": false
}

```

Screenshot:



3. Get User by ID

Endpoint:

`GET /api/users/{id}`

Example:

`GET /api/users/1`

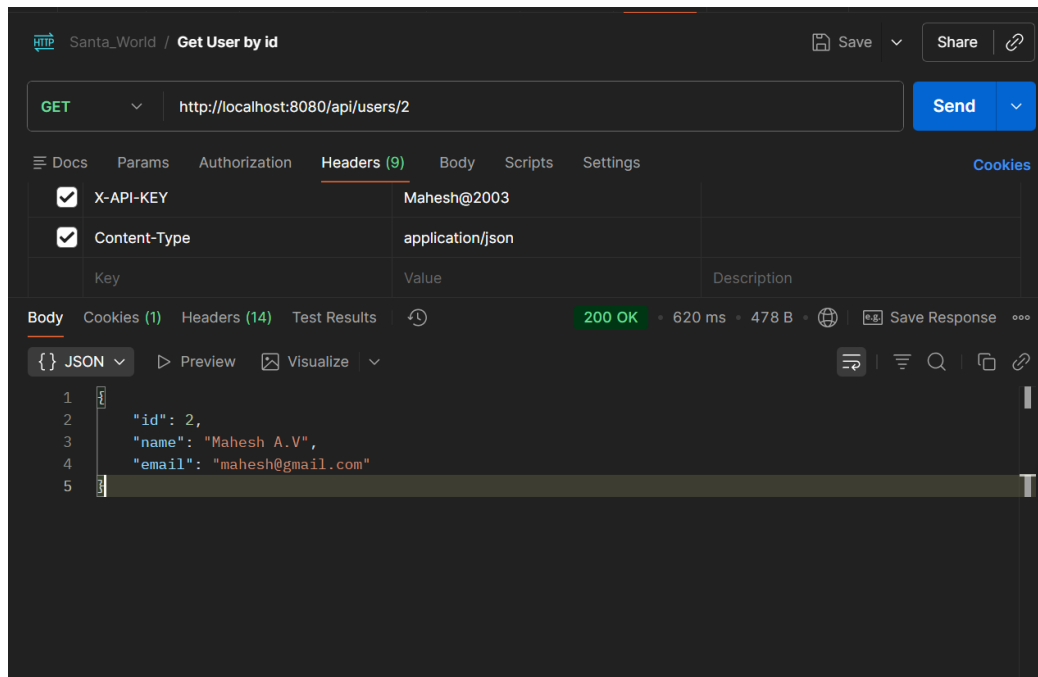
Response:

```

{
  "id": 2,
  "name": "Mahesh A.V",
  "email": "mahesh@gmail.com"
}

```

Screenshot:



Task APIs

4. Create Task

Endpoint:

POST `/api/tasks`

Request Body:

```
{
  "title": "Complete task management",
  "description": "Task Management system by Sentra World",
  "status": "DONE",
  "priority": "MEDIUM",
  "dueDate": "2026-01-30",
  "assignedToUserId": 1
}
```

Response:

```
{
  "id": 2,
  "title": "Complete task management",
  "description": "Task Management system by Sentra World",
  "status": "DONE",
  "priority": "MEDIUM",
  "dueDate": "2026-01-30",
  "createdAt": "2026-01-10T00:46:33",
  "updatedAt": "2026-01-10T00:46:33",
  "assignedTo": {
    "id": 2,
    "name": "Mahesh A.V",

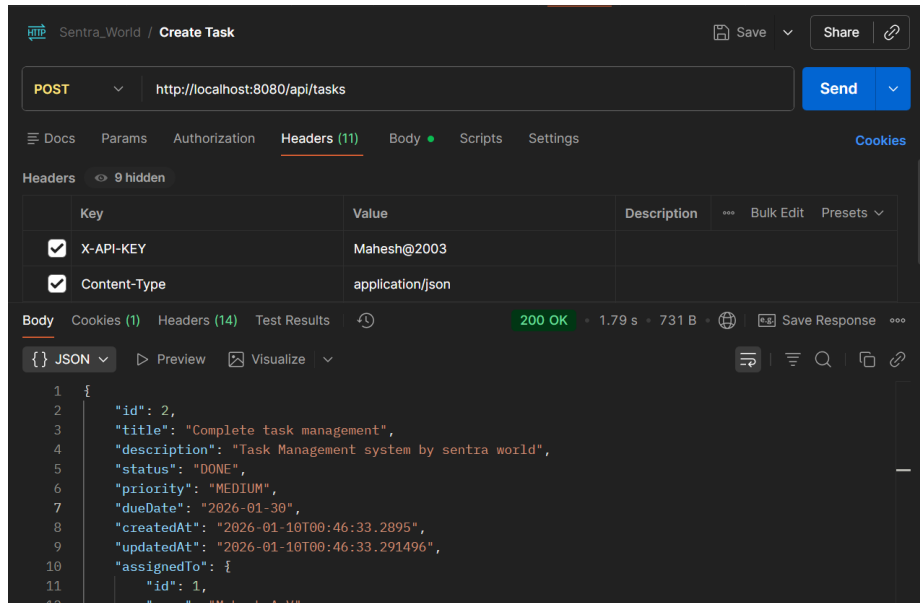
```

```

    "email": "mahesh@gmail.com"
  }
}

```

Screenshot:



5. Get All Tasks (Filters & Pagination)

- Get All Tasks

Endpoint:

GET /api/tasks

Response:-

```

{
  "content": [
    {
      "id": 3,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
      "createdAt": "2026-01-10T02:03:16.650458",
      "updatedAt": "2026-01-10T02:03:16.652496",
      "assignedTo": {
        "id": 1,
        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
      }
    },
    {
      "id": 4,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "HIGH",
      "dueDate": "2026-02-01",

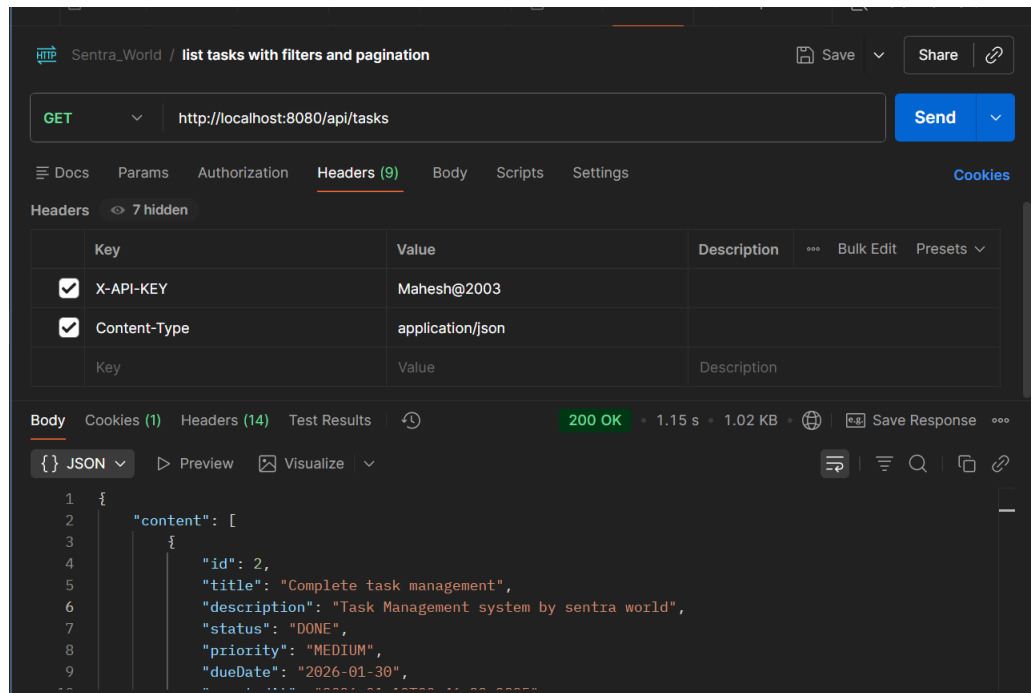
```

```

        "createdAt": "2026-01-10T02:06:28.589008",
        "updatedAt": "2026-01-10T02:06:28.591011",
        "assignedTo": {
            "id": 1,
            "name": "Mahesh A.V",
            "email": "mahesh@example.com"
        }
    },
    "pageable": {
        "pageNumber": 0,
        "pageSize": 10,
        "sort": {
            "empty": true,
            "sorted": false,
            "unsorted": true
        },
        "offset": 0,
        "paged": true,
        "unpaged": false
    },
    "totalPages": 1,
    "totalElements": 2,
    "last": true,
    "size": 10,
    "number": 0,
    "sort": {
        "empty": true,
        "sorted": false,
        "unsorted": true
    },
    "first": true,
    "numberOfElements": 2,
    "empty": false
}

```

Screenshot:-



- **Pagination**

Endpoint:

GET /api/tasks?page=0&size=5

Response:

```

{
  "content": [
    {
      "id": 2,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
      "createdAt": "2026-01-10T02:03:16.650458",
      "updatedAt": "2026-01-10T02:03:16.652496",
      "assignedTo": {
        "id": 1,
        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
      }
    },
    {
      "id": 3,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "HIGH",
      "dueDate": "2026-02-01",
      "createdAt": "2026-01-10T02:06:28.589008",
      "updatedAt": "2026-01-10T02:06:28.591011",
      "assignedTo": {
        "id": 1,
        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
      }
    }
  ]
}

```

```

        "email": "mahesh@example.com"
      }
    },
    "pageable": {
      "pageNumber": 0,
      "pageSize": 5,
      "sort": {
        "empty": true,
        "sorted": false,
        "unsorted": true
      },
      "offset": 0,
      "paged": true,
      "unpaged": false
    },
    "totalPages": 1,
    "totalElements": 2,
    "last": true,
    "size": 5,
    "number": 0,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "first": true,
    "numberOfElements": 2,
    "empty": false
  }
}

```

Screenshot:

The screenshot shows a REST client interface with the following details:

- Request:** GET `http://localhost:8080/api/tasks?page=0&size=5`
- Headers:**
 - X-API-KEY: Maresh@2003
 - Content-Type: application/json
- Response:** 200 OK, 2.19 s, 1.02 KB
- Body (JSON):**

```

{
  "content": [
    {
      "id": 2,
      "title": "Complete task management",
      "description": "Task Management system by sentia world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
    }
  ]
}

```

- **Filter by Status**

Endpoint:

GET /api/tasks?status=DONE

Response:

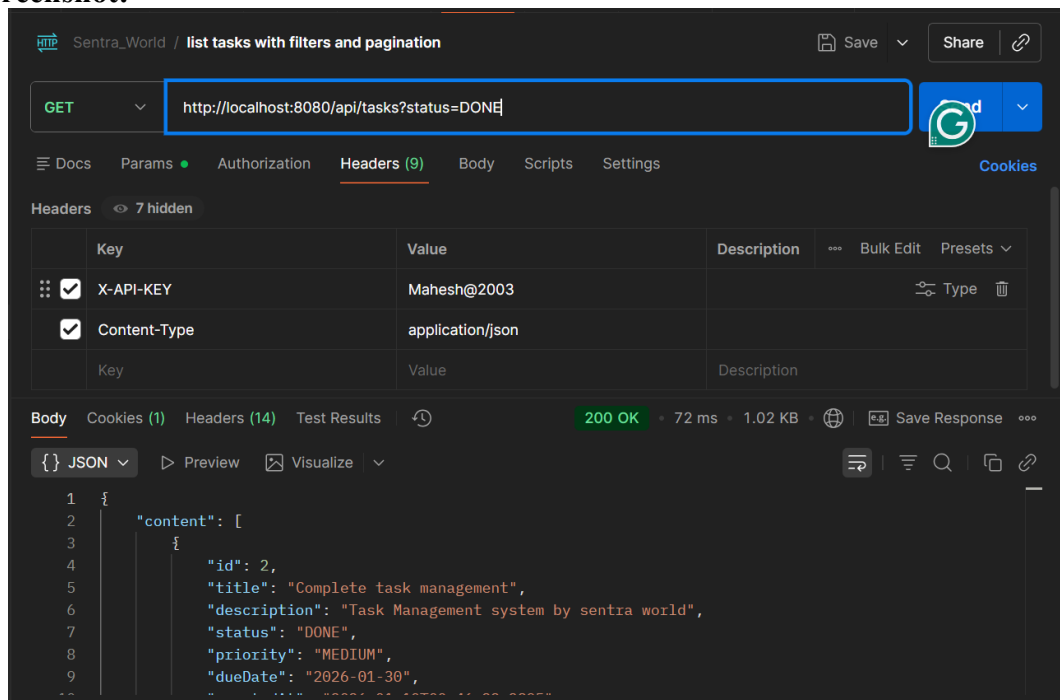
```
{
  "content": [
    {
      "id": 2,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
      "createdAt": "2026-01-10T02:03:16.650458",
      "updatedAt": "2026-01-10T02:03:16.652496",
      "assignedTo": {
        "id": 1,
        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
      }
    },
    {
      "id": 3,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "HIGH",
      "dueDate": "2026-02-01",
      "createdAt": "2026-01-10T02:06:28.589008",
      "updatedAt": "2026-01-10T02:06:28.591011",
      "assignedTo": {
        "id": 1,
        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
      }
    }
  ],
  "pageable": {
    "pageNumber": 0,
    "pageSize": 10,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "offset": 0,
    "paged": true,
    "unpaged": false
  },
  "totalPages": 1,
  "totalElements": 2,
}
```

```

    "last": true,
    "size": 10,
    "number": 0,
    "sort": {
      "empty": true,
      "sorted": false,
      "unsorted": true
    },
    "first": true,
    "numberOfElements": 2,
    "empty": false
  }
}

```

Screenshot:



- **Filter by Priority**

Endpoint:

GET /api/tasks?priority=MEDIUM

Response:

```

{
  "content": [
    {
      "id": 2,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
      "createdAt": "2026-01-10T02:03:16.650458",
      "updatedAt": "2026-01-10T02:03:16.652496",
      "assignedTo": {
        "id": 1,

```

```

        "name": "Mahesh A.V",
        "email": "mahesh@example.com"
    }
},
],
"pageable": {
    "pageNumber": 0,
    "pageSize": 10,
    "sort": {
        "empty": true,
        "sorted": false,
        "unsorted": true
    },
    "offset": 0,
    "paged": true,
    "unpaged": false
},
"totalPages": 1,
"totalElements": 1,
"last": true,
"size": 10,
"number": 0,
"sort": {
    "empty": true,
    "sorted": false,
    "unsorted": true
},
"first": true,
"numberOfElements": 1,
"empty": false
}

```

Screenshot:

The screenshot displays a REST client interface with the following details:

- URL:** `http://localhost:8080/api/tasks?priority=MEDIUM`
- Method:** GET
- Status:** 200 OK (561 ms, 1.02 KB)
- Headers:**

Key	Value	Description
X-API-KEY	Mahesh@2003	
Content-Type	application/json	
- Body (JSON):**

```

{
  "content": [
    {
      "id": 2,
      "title": "Complete task management",
      "description": "Task Management system by sentra world",
      "status": "DONE",
      "priority": "MEDIUM",
      "dueDate": "2026-01-30",
      "createdAt": "2026-01-10T00:46:33.2895",
    }
  ]
}

```

6. Get Task by ID

Endpoint:

GET /api/tasks/{id}

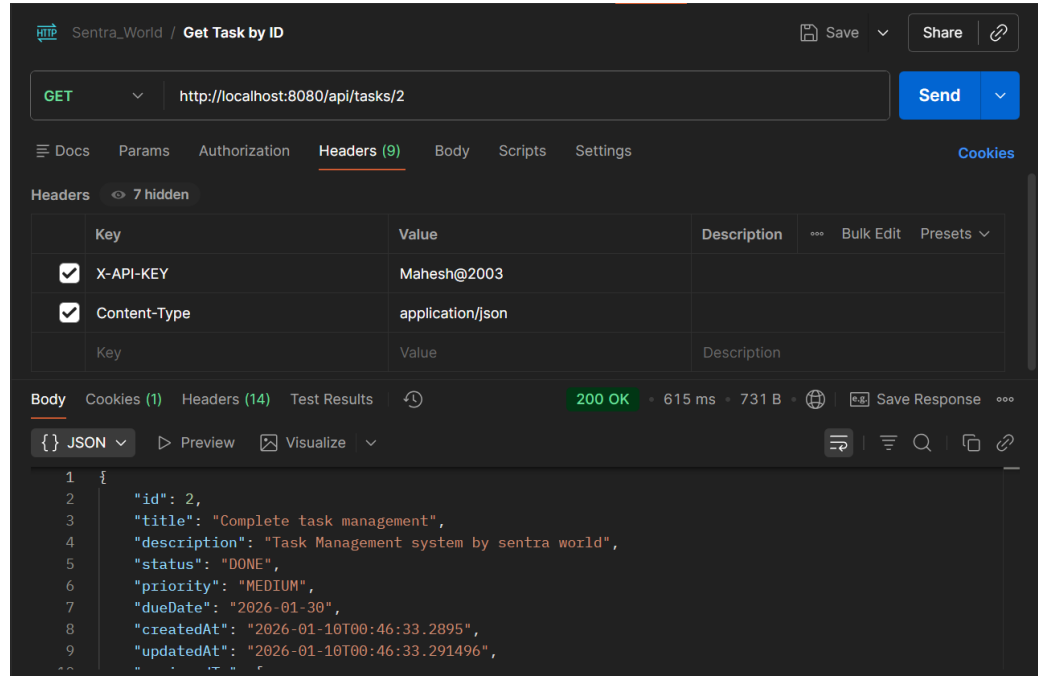
Example:

GET /api/tasks/2

Response:

```
{
  "id": 2,
  "title": "Complete task management",
  "description": "Task Management system by sentra world",
  "status": "DONE",
  "priority": "MEDIUM",
  "dueDate": "2026-01-30",
  "createdAt": "2026-01-10T02:03:16.650458",
  "updatedAt": "2026-01-10T02:03:16.652496",
  "assignedTo": {
    "id": 1,
    "name": "Mahesh A.V",
    "email": "mahesh@example.com"
  }
}
```

Screenshot:



7. Update Entire Task (PUT)

Endpoint:

PUT /api/tasks/{id}

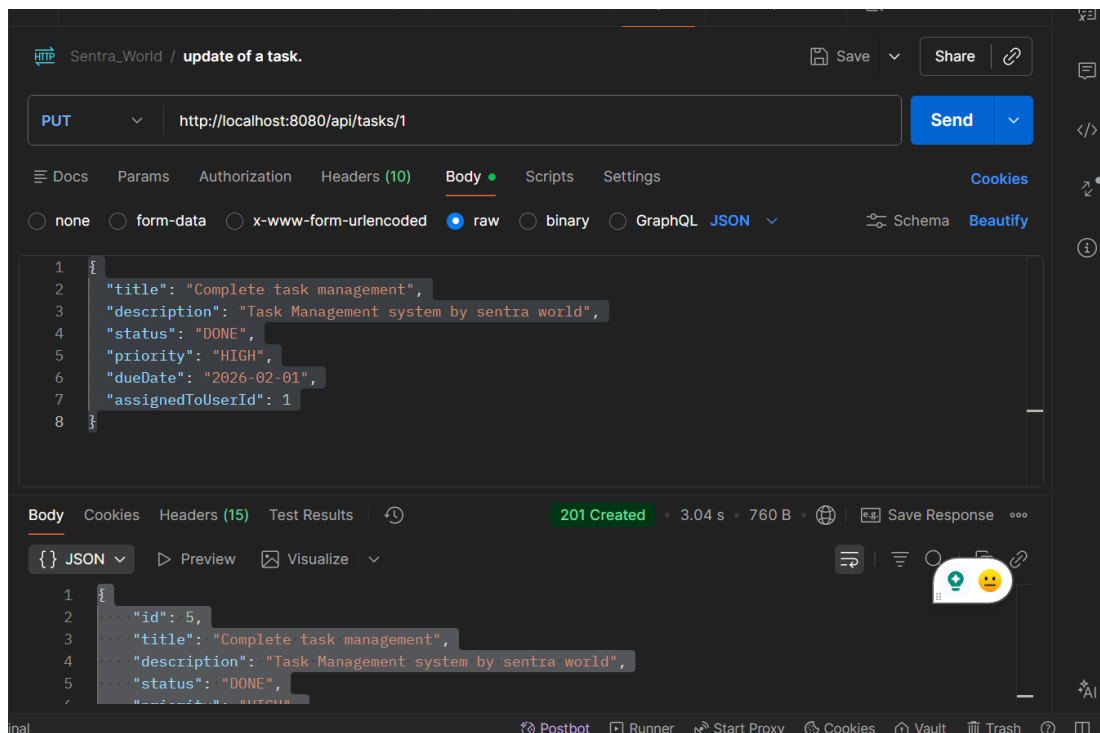
Request Body:

```
{
  "title": "Complete task management",
  "description": "Task Management system by Sentra World",
  "status": "DONE",
  "priority": "HIGH",
  "dueDate": "2026-02-01",
  "assignedToUserId": 1
}
```

Response:

```
{
  "id": 5,
  "title": "Complete task management",
  "description": "Task Management system by sentra world",
  "status": "DONE",
  "priority": "HIGH",
  "dueDate": "2026-02-01",
  "createdAt": "2026-01-10T10:47:07.527704",
  "updatedAt": "2026-01-10T10:47:07.527704",
  "assignedTo": {
    "id": 1,
    "name": "Mahesh A.V",
    "email": "mahesh@example.com"
  }
}
```

Screenshot:



8. Update Task Status (PATCH)

Purpose:

Update **only one field** (status) instead of the full task.

Endpoint:

PATCH /api/tasks/{id}/status?status=DONE

Example:

PATCH /api/tasks/3/status?status=DONE

Response:

```
{
  "id": 3,
  "title": "Complete task management",
  "description": "Task Management system by sentra world",
  "status": "TODO",
  "priority": "MEDIUM",
  "dueDate": "2026-01-30",
  "createdAt": "2026-01-10T02:03:16.650458",
  "updatedAt": "2026-01-10T10:50:53.677619",
  "assignedTo": {
    "id": 1,
    "name": "Mahesh A.V",
    "email": "mahesh@example.com"
  }
}
```

9. Delete Task

Endpoint:

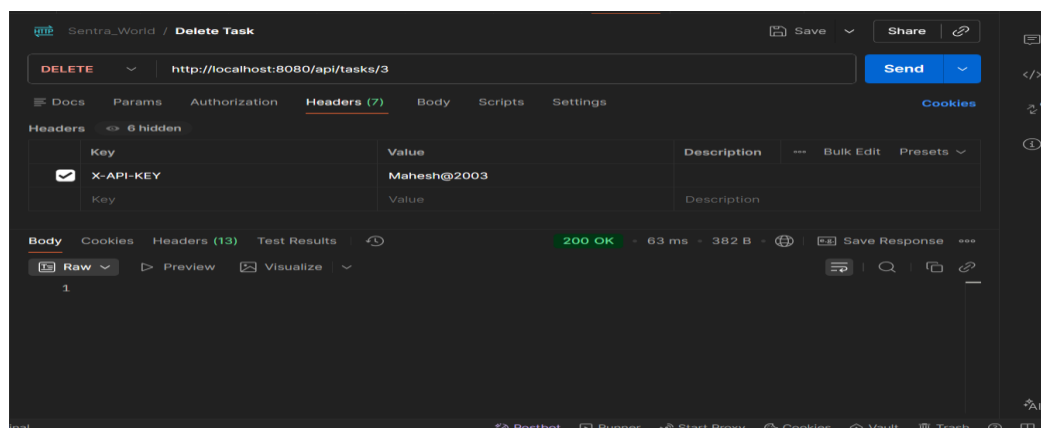
DELETE /api/tasks/{id}

Example:

DELETE /api/tasks/3

Response:

200 No Content

Screenshot:

Error Responses

I. Resource Not Found (404)

```
{  
  "message": "Task not found",  
  "timestamp": "2026-01-10T01:28:43",  
  "status": 404  
}
```

II. Bad Request (400)

```
{  
  "message": "Invalid request data",  
  "status": 400  
}
```

III. Unauthorized (401)

```
{  
  "message": "Invalid API Key",  
  "status": 401  
}
```

Testing Tool

- Postman
- Add X-API-KEY in Headers
- Use JSON body
- Content-Type: application/json

Conclusion

This Task Management API supports:

- User management
- Task CRUD operations
- Filtering & pagination
- Secure API-key-based access
- REST-compliant HTTP methods